YEAR 7 MATHS · CHAPTER 10 · MIND MAP

Measurement

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1 Metric units of length

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm.

2.5 m = **250 cm**

3400 m = **3.4 km**

45 mm = **4.5 cm**

Big unit to small: multiply. Small unit to big: divide.

2 Perimeter

Add all the sides.

Rectangle 7 by 4: $2(7+4) = 22$

Square side 5: $4 \times 5 = 20$

Regular hexagon, side 3: $6 \times 3 = 18$

3 Circles and circumference

$C = \pi d = 2\pi r$. The radius is half the diameter.

$d = 6$: $C = 6\pi \approx 18.8$

$r = 4$: $C = 8\pi \approx 25.1$

$\pi \approx 3.14$

4 Arcs and sectors

Arc length $= \frac{\theta}{360} \times 2\pi r$. Add the straight edges for the perimeter.

Semicircle $r = 5$: arc $5\pi \approx 15.7$, perimeter $10 + 5\pi \approx 25.7$

Quarter circle $r = 4$: arc $2\pi \approx 6.3$, perimeter $8 + 2\pi \approx 14.3$

5 Area of rectangles

$A = lw$, in square units.

$8 \times 3 = 24$ cm²

1 cm² = 100 mm², 1 m² = 10 000 cm²

Square, side 5: 25 cm²

2 m by 3 m: 6 m²

Watch out: to convert area units, square the length conversion.

6 Area of parallelograms

$A = bh$, with the height at right angles to the base.

Base 6, height 4: $A = 24$

Base 8 cm, height 5 cm: $A = 40$ cm²

Watch out: use the perpendicular height, not the slanted side.

7 Area of triangles

$A = \frac{1}{2}bh$: half a parallelogram.

Base 10, height 3: $A = 15$

Base 6, height 7: $A = 21$

Watch out: don't forget the $\frac{1}{2}$.

8 Composite shapes

Split into shapes you know and add, or take a shape away.

L shape: $8 \times 3 + 4 \times 2 = 24 + 8 = 32$

10 by 6 with a 2 by 2 hole: $60 - 4 = 56$

9 Volume of rectangular prisms

$V = lwh$, in cubic units.

$5 \times 3 \times 2 = 30$ cm³

Cube, side 3: $3 \times 3 \times 3 = 27$ cm³

10 Volume of triangular prisms

$V = Ah$: area of the triangle times the length.

Triangle $\frac{1}{2} \times 4 \times 3 = 6$, length 10: $V = 60$

Triangle base 6, height 5: area 15. Length 4: $V = 60$

11 Capacity

1 L = 1000 mL, 1 cm³ = 1 mL, 1 m³ = 1000 L.

250 mL = **0.25 L**

A 10 cm cube holds 1000 cm³, or **1 L**

1.5 L = **1500 mL**

12 Mass and temperature

1 t = 1000 kg, 1 kg = 1000 g.

2.5 kg = **2500 g**

Water freezes at 0°C and boils at 100°C

750 g = **0.75 kg**

From $-2°$C to $5°$C is a rise of $7°$